

Loops and Enums

Condition-controlled Loops

while loops

```
while (Condition)
{
    //checks at each run
    //if the condition is true
}
```

```
do
{
    //runs the code inside the do
    //statement at least once, then
    //continues like the regular while
    //loop (when the condition is true)
}
while (Condition);
```

Loops with a counter (for-loop)

```
for (Counter Variable; Condition; Counter Increment/Decrement)
{
    //Do something with each iteration
}
```

- incremental for-loop

```
for (int i = 0; i < duration; i++)
{
    //Do something with each iteration
}
```

- decremental for-loop

```
for (int i = duration; i >= 0; i--)
{
    //Do something with each iteration
}
```

Enumerate array with ..

```
int[] numbers = { 1, 2, 3, 4, 5, 6, 7, 8, 9 };
```

... a for loop:

```
for (int i = 0; i < numbers.length; i++)  
{  
    Console.WriteLine(numbers[i]);  
}
```

... a foreach loop

```
foreach (int item in numbers)  
{  
    Console.WriteLine(item);  
}
```

Enums

- custom datatype with fixed values
- each value corresponds to an integer

```
enum Weekday { Mon, Tue, Wed, Thu, Fri, Sat, Sun };
```

```
Weekday day = Weekday.Mon;
```

```
if (day == Weekday.Mon)  
{  
    //...  
}
```

switch-Statement

```
string fruit = "Apple";
```

```
switch (fruit)
{
    case "Apple":
        Console.WriteLine("The fruit is an apple");
        break;
    case "Pear":
        Console.WriteLine("The fruit is a pear");
        break;
    case "Cherry":
        Console.WriteLine("The fruit is a cherry");
        break;
    default:
        Console.WriteLine("The fruit is another fruit");
        break;
}
```

```
Weekday day = Weekday.Mon;
```

```
switch (day)
{
    case Weekday.Mon:
    case Weekday.Tue:
    case Weekday.Wed:
    case Weekday.Thu:
    case Weekday.Fri:
        Console.WriteLine("Today is a weekday");
        break;
    case Weekday.Sat:
    case Weekday.Sun:
        Console.WriteLine("Today is the weekend");
        break;
    default:
        Console.WriteLine("Unexpected error");
        break;
}
```